Aarish Muhammad

Islamabad, Pakistan

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Skills

- Python
- Linear Algebra
- Train and Evaluate
- Data Pre Processing
- NLP Basics
- Pandas
- Version Control
- Prompt Engineering
- Databases (SQL)
- Problem Solving
- Team Work
- Creative Thinking

Profile

BS Artificial Intelligence student at NUST Islamabad with hands-on experience in building AI-powered applications and applying machine learning to real-world problems. Worked with Python, Flask, CNN, KNN, Pandas, Scikit-learn, NumPy, Matplotlib, Seaborn, Git, and GitHub, with basic knowledge of model training and evaluation, APIs, backend integration, deployment basics, vector databases, RAG, prompt engineering, and generative AI. Focused on creating practical and impactful AI solutions.

Education

Matric: Muslim Hands School of Excellence

2021 - 2023

Passed with A+ grade scoring 94% marks

FSC (Pre-Eng): Punjab College of Science, Blue Area

2023 - 2025

Passed with A+ grade scoring 89%% marks

BS (Artificial Intelligence): NUST H-12 Campus, Islamabad

2025 - Ongoing (2nd Semester)

Achieved 3.71 GPA in 1st Semester

Professional Experience

Internship at Arch Technologies (Remote): 3 Months

Hands on Work with Projects relating to AI/ML field; developed tools like an Email Spam Classifier and a Digit Recognizer using various ML models.

Projects

All Projects at: github.com/aarish-ai

Digit Recognition — MNIST : TensorFlow, CNN, scikit-learn

Built an end-to-end handwritten digit recognition pipeline using a Convolutional Neural Network trained on the MNIST dataset.
The project preprocesses image data, trains and evaluates the model, and saves predictions with performance visualizations and ~99% test accuracy.
What I learned: CNN architecture, image preprocessing, model training/evaluation, data augmentation, and saving reusable ML models.

Email Spam Classification : Python, scikit-learn, NLTK

Developed a text classification pipeline to identify Emails as spam or ham using TF-IDF feature extraction and Multinomial Naive Bayes.
The system handles text cleaning, feature engineering, model training, and evaluation, achieving around 97% accuracy on the test set.
What I learned: NLP preprocessing, TF-IDF, Naive Bayes classification, and building a complete ML workflow for text data.

Chess — Play Chess Against AI : Flask, JavaScript, HTML/CSS

Built a browser-based chess application that allows users to play against a computer-controlled opponent through an interactive web interface.
The project integrates frontend game logic with a Flask backend to create a lightweight and responsive gameplay experience.
What I learned: Full-stack integration, backend routing with Flask, frontend interaction, and implementing game logic in JavaScript.

NUSTcord : Java, Servlets/JSP, H2 Database

Developed a structured communication platform inspired by Discord with secure authentication, friend networking, servers, channels, and messaging.
The application follows a layered MVC architecture and includes role-based access, password hashing, session management, and an admin dashboard.
What I learned: Java web development, MVC architecture, database design, secure authentication, and modular software structure.

Truth Mirror (Ongoing) : Python, APIs, Gemini/LLMs

Designed an evidence-based claim verification MVP that analyzes natural-language claims and returns structured verdicts with citations and uncertainty handling.
The system decomposes claims, retrieves evidence from multiple sources, ranks credibility, and produces supported or contradicted conclusions.
What I learned: Retrieval-based systems, claim decomposition, evidence ranking, API integration, and building LLM-assisted verification pipelines.

Courses

Agentic AI by Andrew Ng — deeplearning.ai (Ongoing)

Focused on the concepts and application of agentic systems, workflow orchestration, and how AI agents can reason, act, and use tools effectively.
The course also explores practical applications of LLM-based agents for building intelligent, task-oriented AI systems.

Python for AI and Data Science — NUST Islamabad

Covered the use of Python for artificial intelligence and data science tasks, including data handling, analysis, and basic machine learning workflows.
The course strengthened core programming skills while introducing practical tools and techniques used in AI development.

Python Programming — NUML Islamabad

Taught the fundamentals of Python programming, including syntax, control structures, functions, and problem-solving logic.
It built a strong foundation in writing clean and efficient code for general programming and future technical work.

Python Beginner Course by Mosh Hamedani — YouTube

Introduced Python from the ground up in a simple and structured way, covering the language basics and core programming concepts.
The course helped reinforce beginner-level understanding through clear explanations and practical coding examples.